EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 3

PRINTABLE PRACTICE WORKSHEET

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Candidate Name:		**Roll No.:**	
Class / Grade:		**Date:**	
School / Institution:		**Test Centre:**	

Subject	Biology	**Topic**	Biology Comprehensive Practice Test 3	**Total Marks**	100
Total Questions	45	**Time Allowed**	1 Hour (60 minutes)	**Version**	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	**45**	**—**	**100**

Section A — Multiple Choice — 30 marks

1. Which organelle contains most of a eukaryotic cell's DNA? — Basic · 2 pts

A. Ribosome **B.** Lysosome
C. Golgi apparatus **D.** Nucleus

Answer: ☐A ☐B ☐C ☐D

2. Ribosomes are directly involved in: — Basic · 2 pts

A. Protein synthesis **B.** Storage of the cell's DNA
C. Storage of lipids only **D.** Digestion of the cell wall

Answer: ☐A ☐B ☐C ☐D

3. The plasma membrane is mainly a: — Basic · 2 pts

A. Rigid wall made of cellulose in all cells **B.** Layer made only of DNA
C. Selective boundary containing a lipid bilayer and proteins **D.** Barrier that contains no protein

Answer: ☐A ☐B ☐C ☐D

4. Which molecule is a polymer made from amino acids? — Basic · 2 pts

A. Starch **B.** Protein
C. DNA **D.** Triglyceride

Answer: ☐A ☐B ☐C ☐D

5. Enzymes accelerate reactions primarily by: **Basic · 2 pts**

A. Lowering the activation energy of the reaction
B. Increasing the overall energy change of the reaction
C. Being used up as one of the reactants
D. Raising the temperature of every reaction

Answer: ☐A ☐B ☐C ☐D

6. A very high temperature can reduce enzyme activity because: **Intermediate · 2 pts**

A. All substrates disappear from the solution
B. ATP is converted into DNA
C. Water is converted into a carbohydrate
D. Protein structure can be disrupted, changing the active site

Answer: ☐A ☐B ☐C ☐D

7. The iodine test is commonly used to detect: **Intermediate · 2 pts**

A. Protein
B. Lipids
C. Starch
D. Vitamin C

Answer: ☐A ☐B ☐C ☐D

8. A positive Benedict's test after heating can indicate: **Intermediate · 2 pts**

A. Starch
B. Reducing sugars
C. Proteins
D. Nucleic acids

Answer: ☐A ☐B ☐C ☐D

9. The Biuret test is associated with: **Intermediate · 2 pts**

A. Starch granules
B. Lipid droplets
C. Mineral ions
D. Peptide bonds in proteins

Answer: ☐A ☐B ☐C ☐D

10. The emulsion test is commonly used for: **Intermediate · 2 pts**

A. Lipids
B. Starch
C. Protein
D. Reducing sugars

Answer: ☐A ☐B ☐C ☐D

11. If an enzyme's active site changes shape, the most direct consequence is likely: **Advanced · 2 pts**

A. An immediate increase in all reactions
B. Creation of a new organelle
C. Reduced ability to bind or convert its substrate
D. Loss of all the cell's DNA

Answer: ☐A ☐B ☐C ☐D

12. Which statement best describes hydrolysis? **Advanced · 2 pts**

A. Two monomers always join together without water
B. A bond is broken using water, often splitting a larger molecule
C. A protein is converted into DNA
D. Energy is never involved in the process

Answer: ☐A ☐B ☐C ☐D

13. Why are controls important in food tests? **Advanced · 2 pts**

A. They help show that observed colour changes are linked to the sample and the test procedure
B. They make every sample give a positive result
C. They remove the need to use reagents
D. They prove the exact concentration of the substance

Answer: ☐A ☐B ☐C ☐D

14. A positive qualitative food test generally tells you that: **Advanced · 2 pts**

A. The exact concentration of the substance is known
B. The sample is healthy to eat
C. Every related molecule is present in the sample
D. The target substance or reactive group is detected under the test conditions

Answer: ☐A ☐B ☐C ☐D

15. Which statement best connects structure and function? **Advanced · 2 pts**

A. Molecular structure never affects function
B. Only the size of a molecule decides its function
C. Molecular structure influences interactions, so changing structure can alter function
D. Function is independent of molecular shape

Answer: ☐A ☐B ☐C ☐D

Section B — True / False — 20 marks

16. All proteins are enzymes. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

17. Enzymes are generally unchanged at the end of the reactions they catalyse. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

18. A negative iodine test proves that a food contains no carbohydrates of any type. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

19. The cell membrane helps maintain different conditions inside and outside the cell. Basic · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

20. DNA and RNA are nucleic acids. Basic · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

21. Lipids can contribute to membranes and energy storage. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

22. A qualitative test always gives an exact concentration. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

23. Substrate concentration can affect enzyme reaction rate under suitable conditions. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

24. Extreme pH can alter the structure and activity of many enzymes. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

25. Controls are unnecessary when comparing two food samples. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

Section C — Fill in the Blank 20 marks

26. The monomers that form proteins are _____ acids. Intermediate · 2 pts
Answer: _____

27. The enzyme region where a substrate binds is the _____ site. Intermediate · 2 pts
Answer: _____

28. The main carbohydrate storage polymer in plants is _____. Intermediate · 2 pts
Answer: _____

29. Enzymes are biological _____ because they speed up reactions without being used up. Intermediate · 2 pts
Answer: _____

30. Glucose and fructose are examples of simple sugars called _____. Intermediate · 2 pts
Answer: _____

31. The permanent change in an enzyme's shape caused by high temperature or extreme pH is called _____. Advanced · 2 pts
Answer: _____

32. Fats and oils belong to a group of biological molecules called _____. Advanced · 2 pts
Answer: _____

33. The reaction that joins two molecules and releases water is called a _____ reaction. Advanced · 2 pts
Answer: _____

34. The organelle that releases most of a cell's usable energy as ATP is the _____. Advanced · 2 pts
Answer: _____

35. The small units that join to form a nucleic acid such as DNA are called _____. Advanced · 2 pts
Answer: _____

Section D — Column Matching — 10 marks

36–40. Match each term in Column A with the correct description in Column B. **Advanced · 10 pts**

Column A	Answer	Column B
Carbohydrate	____	**A.** Membranes and energy storage
Lipid	____	**B.** Starch detection
Protein	____	**C.** Protein detection
Iodine test	____	**D.** Energy supply or structural roles
Biuret test	____	**E.** Structure, transport, catalysis and signalling

Section E — Short Answer — 20 marks

41. Explain why enzyme activity depends on three-dimensional protein structure. **Advanced · 4 pts**

42. A food gives a positive Benedict's test but a negative iodine test. What can reasonably be concluded? **Advanced · 4 pts**

43. Why is a control important when carrying out a biochemical food test? **Advanced · 4 pts**

44. Distinguish hydrolysis from condensation using biological molecules as examples. **Advanced · 4 pts**

45. Explain how membrane structure supports selective transport. **Advanced · 4 pts**

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: / 100	Checked By:
Result / Grade:	Date Checked:

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Biology Comprehensive Practice Test 3 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

1. D
2. A
3. C
4. B
5. A
6. D
7. C
8. B
9. D
10. A
11. C
12. B
13. A
14. D
15. C

Section B — True / False

16. False
17. True
18. False
19. True
20. True
21. True
22. False
23. True
24. True
25. False

Section C — Fill in the Blank (accepted answers)

26. amino
27. active
28. starch
29. catalysts / catalyst
30. monosaccharides / monosaccharide
31. denaturation / denaturing / denatured
32. lipids / lipid
33. condensation / dehydration synthesis
34. mitochondrion / mitochondria
35. nucleotides / nucleotide

Section D — Column Matching

36. Carbohydrate — D
37. Lipid — A
38. Protein — E
39. Iodine test — B
40. Biuret test — C

Section E — Short Answer (Model Answers)

41. An enzyme's amino-acid chain folds into a specific three-dimensional shape that forms the active site. The substrate fits into this site, so the enzyme can catalyse the reaction. If heat or extreme pH changes the shape (denaturation), the substrate no longer fits and activity falls.
Marking: 1 mark: the enzyme has an active site with a specific shape; 1 mark: the substrate binds to the active site (complementary); 1 mark: if the shape changes, the substrate no longer fits and activity falls; 1 mark: the shape comes from the folding of the amino-acid chain.

42. A positive Benedict's test shows that a reducing sugar such as glucose is present, and the negative iodine test shows that starch was not detected. The results do not show whether proteins or lipids are present, and a qualitative test cannot tell the exact amount.
Marking: 1 mark: a reducing sugar is present; 1 mark: starch was not detected; 1 mark: the tests do not show whether other molecules (protein, lipid) are present; 1 mark: the tests are qualitative: they do not show how much.

43. A control gives a result to compare with. A negative control, such as water with the reagent, should show no colour change, and a positive control, such as a known starch or sugar solution, shows that the reagent works. This lets you be confident that any colour change in the food sample is caused by the substance being tested for, not by contamination or a faulty reagent.
Marking: 1 mark: it gives a result to compare with; 1 mark: a negative control (for example water) should show no colour change; 1 mark: a positive control (a known sample) shows that the reagent works; 1 mark: so the result can be trusted as caused by the sample, not contamination.

44. Hydrolysis uses water to break bonds and split a larger molecule into smaller units, for example starch into glucose during digestion. Condensation is the opposite: it joins smaller molecules by forming a bond and releases water, for example amino acids joining to make a protein.
Marking: 1 mark: hydrolysis uses water to break bonds; 1 mark: condensation joins molecules and releases water; 1 mark: gives a correct biological example; 1 mark: they are opposite processes (polymer to monomers, and monomers to polymer).

45. The membrane is a phospholipid bilayer whose water-hating (hydrophobic) core blocks most ions and large or polar molecules, while small non-polar molecules such as oxygen can diffuse across. Channel and carrier proteins in the membrane let specific substances pass, so the cell controls what enters and leaves.
Marking: 1 mark: the membrane is a phospholipid bilayer; 1 mark: the water-hating (hydrophobic) core blocks ions and large or polar molecules; 1 mark: small non-polar molecules such as oxygen can diffuse across; 1 mark: channel or carrier proteins let specific substances pass.